

Attack and defense

MANY SEA CREATURES HAVE WEAPONS to defend themselves from predators or to attack prey. Some produce venom (poison) for defense and often advertise their danger with distinctive markings. Lionfish's stripes may alert enemies to their venomous spines, but being easy to see, they hunt at dusk and during the night when they can still surprise their prey out in the open. Stonefish are armed with venomous spines, too, blending perfectly with their background when waiting on a reef for prey to swim by. Octopuses change color to that of their background. If attacked, the blue-ringed octopus produces blue spots to warn that its bite is poisonous. Disappearing in a cloud of ink is another useful trick used by octopuses, squid, and cuttlefish. Most clams can withdraw their delicate soft parts into their shells, but the gaping file shell's tentacles are a deterrent producing an irritating sticky fluid. But no defense method is foolproof. Even the most venomous jellyfish can be eaten by carnivorous turtles that are immune to their stings.



DEADLY STONEFISH
The stonefish is one of the most deadly creatures in the ocean. A stonefish's venom, which is injected through the sharp spines on its back, causes such intense pain that a person stepping on one may go into shock and die.

Long, dorsal spine with venom glands in grooves

INK SCREEN

Cephalopods, which include cuttlefish, squid, and octopuses, produce a cloud of ink when threatened, to confuse an enemy and allow time for escape. The ink, produced in a gland linked to the gut, is ejected in a blast of water from a tubelike funnel near its head.

Ink cloud forming around cuttlefish

Horny projection above eye

Maerl (a chalky, red seaweed) grows in a thick mass along the stony seabed

Three venomous anal spines

KEEP CLEAR

The striped body of lionfish warns predators that they are dangerous. A predator trying to bite a lionfish may be impaled by one or more of its poisonous spines. If it survives, the predator will remember the danger and leave the lionfish alone in future. Lionfish can swim openly looking for smaller prey with little risk of attack. They live in tropical waters from the Indian to the Pacific Oceans. In spite of being poisonous, they are popular aquarium fish because of their beauty.

Stripes warn predators that lionfish is dangerous

BLUE FOR DANGER

If this octopus becomes irritated, or when it is feeding, blue-ringed spots appear on its skin, warning of its poisonous bite. This octopus is only about the size of a person's hand, but its bite can be fatal. One kind of blue-ringed octopus lives in cool shallow waters around parts of Australia. Others are found in tropical waters.